

13 NOV 2025

Examination **BCA+SEM-3 (NED)**
SUBJECT CODE: 25741

25741-[BCAM207-2C]-OPERATING SYSTEM
Duration: 02:00HRS

[CREDIT: 4]
Marks:50

Q-1 Compare and contrast the various operating system structures such as Monolithic, Layered, Virtual Machine, and Client-Server architecture, and give the benefits and challenges of each structure." 10

OR

Q-1(A) What do you mean by operating system? Give primary functions of operating system. 05

Q-1(B) Explain in brief terms Jobs , Processes and file. 05

Q-2 For the given five processes arriving at time 0, in the order with the length of CPU time in milliseconds: 10

Process	P1	P2	P3	P4
Processing Time	10	29	03	07

Consider the FCFS, SJF and RR (time slice= 10 milliseconds) scheduling algorithms for the above set of process which algorithm would give the minimum average waiting time?

OR

Q-2(A) Explain the necessary condition for deadlock to occur. 05

Q-2(B) What do you mean by threads? Explain why to use Threads 05

Q-3 What do you mean by page replacement. Compare and contrast various page replacement algorithm. 10

OR

Q-3(A) A system with 18-bit address uses 6 bits for page number and next 12 bits for offset. Compute the total number of pages and express the following address according to paging scheme 001011(page number) and 000000111000(offset)? 05

Q-3(B) Explain concept of contiguous memory allocation. 05

Q-4 What do you mean by file? Explain various file access method in brief 10

OR

Q-4A List various directories structure. Explain the advantages of organising file directory structure into a tree structure? 05

Q-4B What do you mean by file? Explain various file operations. 05

Q-5 Answer any Ten 10

1 What is the full name of FAT?

2 When does page fault occur?

A. The page is present in memory. B. The deadlock occurs.
 C. The page does not present in memory. D. The buffering occurs.

3 Banker's algorithm is used?

A. To prevent deadlock B. To deadlock recovery
 C. To solve the deadlock D. None of these

4 What is the full name of the DSM?

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- 5 If the page size increases, the internal fragmentation is also?.
A. Decrease B. Increase C. Remains Constant D. None of These
- 6 Which of the following is a condition that causes deadlock?
A Mutual exclusion B. Hold and wait C. Circular wait D. No preemption
E. All Of These
- 7 _____ stores entries about file and their attributes.
A. File Application Table B. File Allocation Table
C. File Access Table. D. File Format Table
- 8 Give full name of FCFS
- 9 Define segmentation.
- 10 Give Full name of SJF
- 11 Segmentation avoids external memory fragmentation (True/False)
- 12 Virtual Memory Space is always smaller than physical memory space.